checkpoint was promoted, not a certified result: it was measured on three meshes rather than the six the final claim uses, and it is superseded by the single-protocol revalidation of Chapter 7, where the per-stack tracking numbers belong. One protocol point belongs here: the reported policy was obtained under the reference-anchored construction of Section 6.4 with a lower actor learning rate than R and a fixed exploration standard deviation of 0.05, which is read from the run manifest rather than assumed to match.

Figure 6.5 summarises the two non-adaptive actuator paths used as baselines. Both use the same policy interface and PhysX task; only the actuator representation changes.

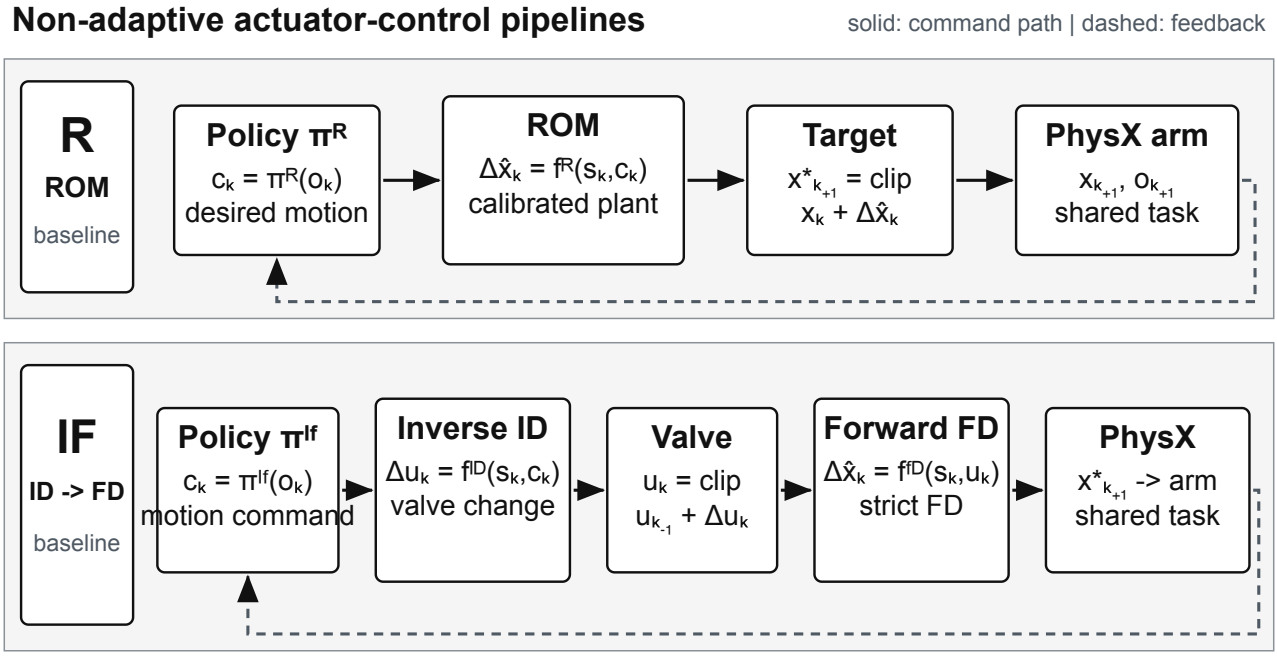


Figure 6.5: The non-adaptive R and IF control pipelines. R sends the policy's desired cylinder motion through the calibrated command ROM. IF composes the independently identified inverse and strict forward models before applying the same bounded cylinder target in PhysX. Both paths use the same observation, action limits and task.

6.7 AF: The Adaptive Stack

AF places online adaptation on the forward model alone. The policy emits the valve command directly, the frozen strict forward model of Section 5.3 converts that command to a predicted extension increment, and PhysX receives the same bounded cylinder target as the other two stacks:

$$\pi_{\text{frozen}} + \text{history residual} \longrightarrow \boldsymbol{u} \longrightarrow f_{\text{FD}} \longrightarrow \text{PhysX}. \tag{6.10}$$

The inverse model is absent from the command path. This is the one structural difference from IF, and it is deliberate: the name of the stack records the two models that remain, AAM and FD. Task, terrain families, meshes, observation contract, reward, gates and the PPO update are inherited unchanged, so the actuator representation and the adaptation path are the only free variables against the baselines.